

Hate Speech Detection Using Machine Learning

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Abstract

Social media platforms contain large amounts of offensive and harmful language. Manual moderation is difficult due to the high volume of posts generated daily. This project investigates how traditional and deep learning models can be used to detect hate speech in tweets. Using the Hate Speech and Offensive Language dataset, several classification models were trained and evaluated, including Logistic Regression, Naive Bayes, Random Forest, Support Vector Machines, Bidirectional LSTM and DistilBERT. Two feature extraction techniques were used for the classical models: Bag-of-Words and TF-IDF. The results show that traditional machine learning models can achieve strong baseline performance, with accuracy close to 0.90. DistilBERT achieved the best performance overall, with an accuracy of 0.9161 and a macro F1 score of 0.7438, demonstrating its ability to capture contextual and semantic meaning and better detect nuanced hate speech. The best classical model, SVM with Bag-of-Words, achieved a macro F1 score of 0.7108, outperforming the Bidirectional LSTM and showing that traditional approaches can remain competitive. However, distinguishing between hate speech and offensive language still remains challenging for all models due to vocabulary overlap and class imbalance.

Introduction

Social media platforms such as Twitter allow users to communicate instantly with a global audience. While this has many benefits, it also enables the spread of harmful and abusive language. Hate speech and offensive language can negatively impact individuals and communities, making automated detection an important task.

Manual moderation is difficult because millions of posts are created every day. Machine learning models can help detect harmful content automatically and support moderation systems.

The goal of this project is to investigate how effectively machine learning models can classify tweets into three categories:

- Hate Speech
- Offensive Language
- Neither

The report is organized as follows. First, the methodology describes the dataset, preprocessing steps, feature extraction techniques, and machine learning models used. Second, the results section presents visualizations and performance comparisons of the models. This is followed by a discussion of the findings and limitations. Finally, the report concludes with a summary of the results and possible future work.

1 Methodology

This section describes the dataset used in the study, the preprocessing steps applied to the text data, and the machine learning techniques used for classification.

1.1 Dataset

The dataset used in this project is the *Hate Speech and Offensive Language Dataset* [1] available on Kaggle. It contains approximately 25,000 tweets labeled into three classes:

- Hate Speech
- Offensive Language
- Neither

The dataset is labeled and therefore suitable for supervised learning. However, the dataset is imbalanced, with most tweets belonging to the offensive language class.

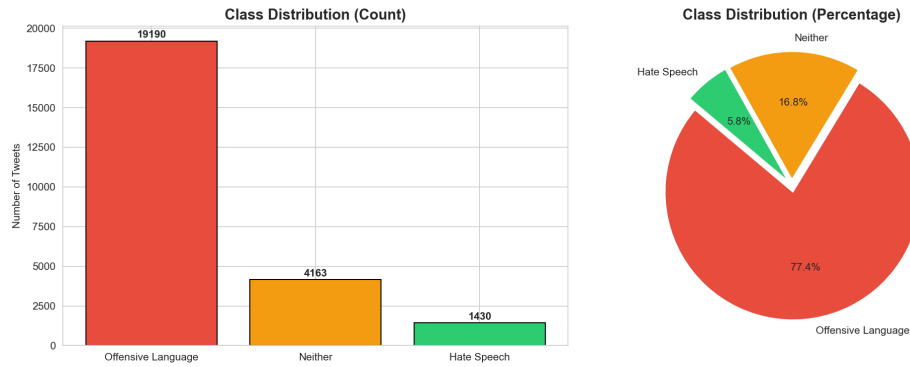


Figure 1: Distribution of tweet classes in the dataset. Offensive language dominates the dataset, while hate speech is the smallest class.

1.2 Text Preprocessing

Before training the machine learning models, the text data was cleaned and standardized. The following preprocessing steps were applied:

- Convert text to lowercase
- Remove URLs and mentions
- Remove special characters
- Tokenization (handled by Bert internally)
- Stopword removal (excluding BERT)
- Lemmatization (excluding LSTM & BERT)

These steps reduce noise in the data and improve model performance.

1.3 Machine Learning Models

Several machine learning algorithms were evaluated:

Classical models

- Logistic Regression
- Multinomial Naive Bayes
- Random Forest
- Support Vector Machine (LinearSVC)

Deep learning models

- Bidirectional LSTM
- DistilBERT

These models are commonly used for text classification and provide a good comparison between probabilistic, linear, and ensemble methods.

1.4 Feature Extraction

Machine learning models cannot directly process text data. Therefore, the tweets were converted into numerical features. Two common techniques were used for the classical models (Logistic Regression, Naive Bayes, Random Forest & SVM):

Bag-of-Words (BoW) represents each document using word frequency counts.

TF-IDF (Term Frequency – Inverse Document Frequency) assigns weights to words based on their importance within the dataset. Words that appear frequently in one document but rarely in others receive higher importance.

TF-IDF is commonly used in text classification tasks because it highlights informative words.

1.5 Deep Learning Model Training

Bidirectional LSTM

The text was tokenized for the LSTM model. The embedding layer was used to transform each word index into a dense vector to help the model capture semantic relationships between words. The Bidirectional LSTM layer was used to read the sentence in both directions: left to right and right to left. This is useful because meaning in language often depends on both previous and following words. Class weights were used by the model to make rare classes more important during training. This helps the model pay more attention to underrepresented classes, which is especially important in hate speech detection. 10% of the training data was used for validation. 5 epochs were used, so that up to 5 full passes could be made through the training data. The model predicted a probability for each class for every test tweet. The class with the highest probability became the predicted class.

DistilBERT

A pretrained DistilBERT model was loaded and adapted for sequence classification and 3 output labels. The model is pre-trained to recognize general language patterns by default. It was fine-tuned on the Kaggle dataset, using Hugging Face Trainer to evaluate the data after each epoch and select the best checkpoint based on macro F1. The model was tested on the same test set as the LSTM and then metrics were calculated.

1.6 Evaluation Metrics

The models were evaluated using several metrics:

- Accuracy
- F1-score (macro)
- F1-score (weighted)
- Confusion matrices

These metrics provide insight into both overall performance and class-specific errors.

2 Results

This section presents the visualizations and performance results obtained from the experiments.

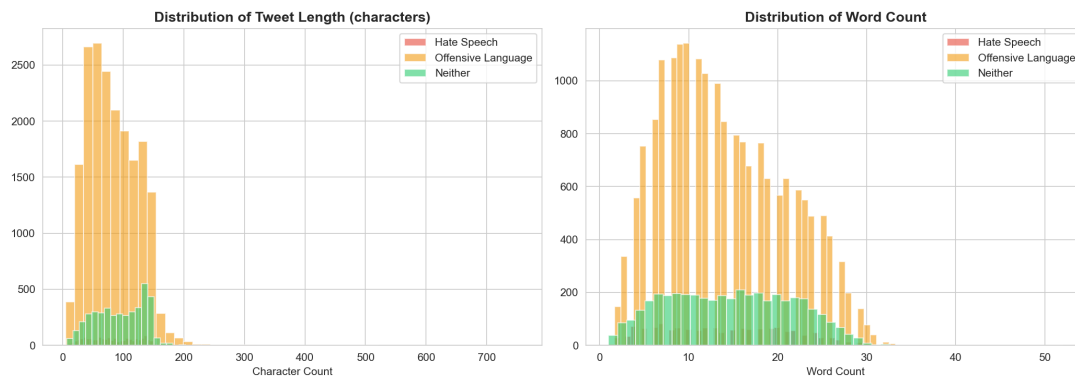


Figure 2: Distribution of tweet length and word count. Most tweets are relatively short.

Figure 2 shows that most tweets contain only a small number of words, which means that models must make predictions using limited textual information.



Figure 3: Word clouds for the three classes. Hate speech and offensive language contain many insults and slurs, while neutral tweets contain more general vocabulary.

The word clouds illustrate the most common words in each category and highlight the differences between the classes.

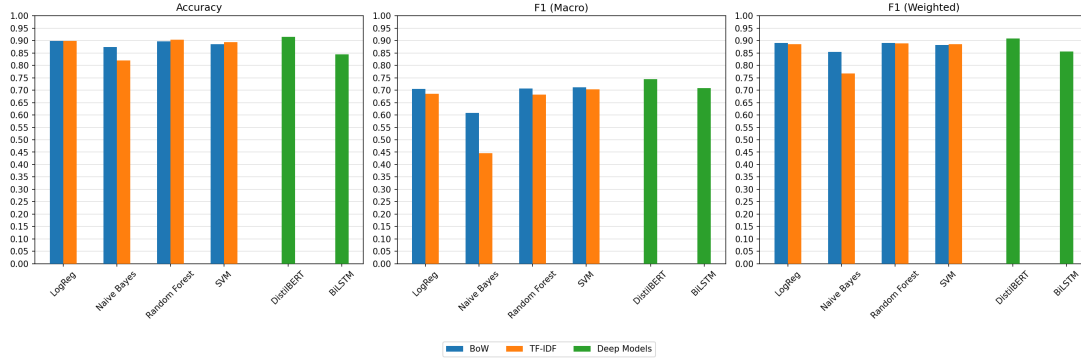


Figure 4: Comparison of model performance.

Model	Features	Accuracy	F1 (Macro)	F1 (Weighted)
DistilBERT	–	0.9161	0.7438	0.9091
SVM (LinearSVC)	BoW	0.8864	0.7108	0.8832
Bidirectional LSTM	–	0.8447	0.7083	0.8571
Random Forest	BoW	0.8977	0.7072	0.8900
Logistic Regression	BoW	0.8989	0.7046	0.8905
SVM (LinearSVC)	TF-IDF	0.8941	0.7040	0.8853
Logistic Regression	TF-IDF	0.8995	0.6865	0.8860
Random Forest	TF-IDF	0.9027	0.6834	0.8886
Naive Bayes	BoW	0.8737	0.6091	0.8540
Naive Bayes	TF-IDF	0.8200	0.4450	0.7674

Table 1: Model Performance Comparison (sorted by F1 Macro)

The results show that TF-IDF does not consistently outperform Bag-of-Words but performance depends on the model and metric. DistilBERT shows the best overall performance with the highest accuracy and F1 scores. LSTM accuracy is approximately 7 percentage points lower than DistilBERT. It shows that it learns patterns, but not as well as transformers.

Notably, SVM (LinearSVC) with Bag-of-Words ranks as the second-best overall model in terms of F1 (Macro) score, outperforming the Bidirectional LSTM, which highlights that traditional approaches can remain highly competitive with deep learning methods. Random Forest (TF-IDF) scores the highest accuracy among classical models (0.9027) and SVM (BoW).

Naive Bayes is the weakest performing model. It struggles with complex language patterns and class imbalance.

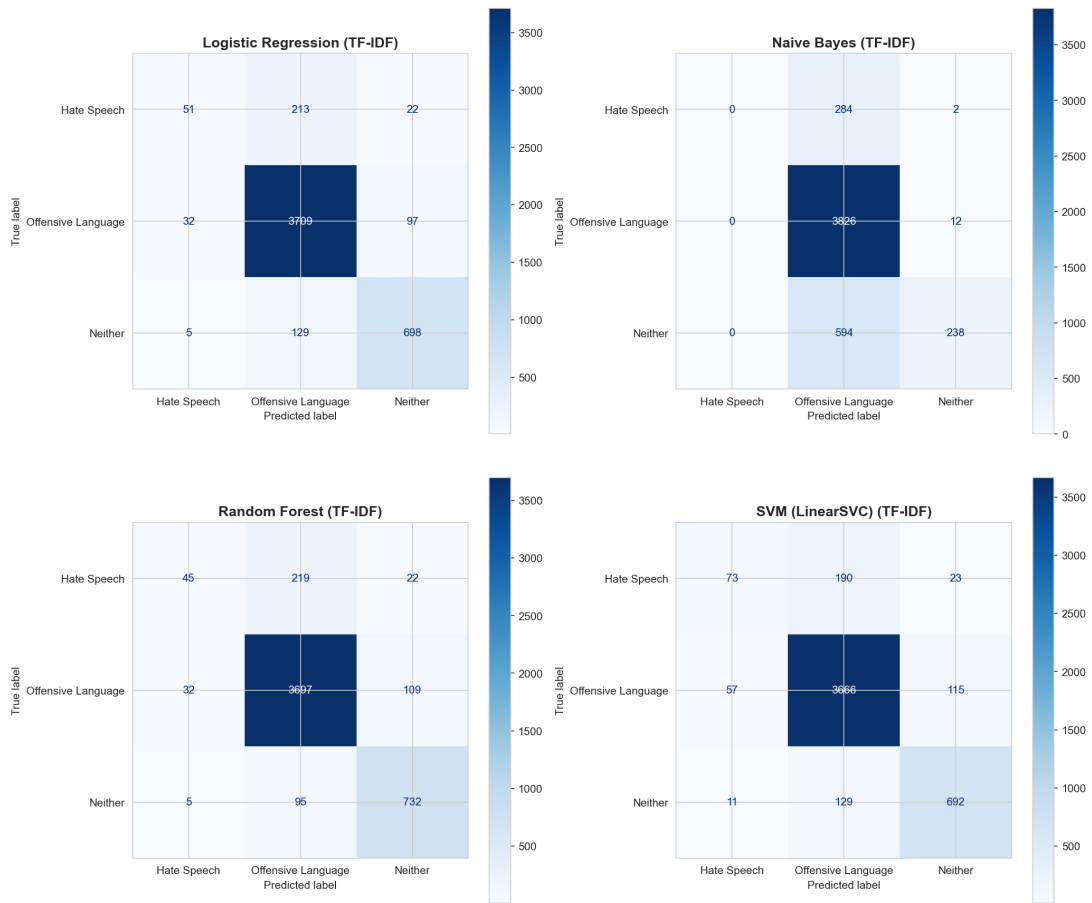


Figure 5: Confusion matrices showing classification performance for the classical models using TF-IDF.

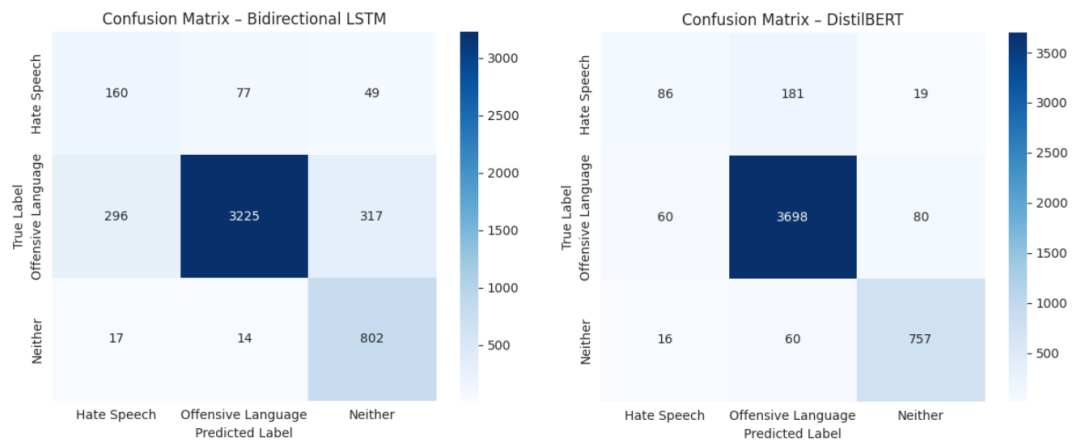


Figure 6: Confusion matrices showing classification performance of the deep learning models.

The confusion matrices reveal that the main difficulty lies in distinguishing between hate speech and offensive language. All models show the same pattern:

- Very strong performance on “Offensive Language”
- Weak performance on “Hate Speech”
- Decent performance on “Neither”

This shows that the models are biased toward the majority class (Offensive Language). All models struggle to correctly identify hate speech and tend to classify it as offensive language. The ‘Neither’ class is reasonably well predicted, but there is still confusion with offensive language. Naïve Bayes performs the worst, Logistic Regression

and Random Forest have similar behavior with slight ability to detect Hate Speech, but still weak. SVM is the best classical model for balance. It is slightly better at detecting Hate Speech.

DistilBERT is more precise and makes fewer mistakes across all classes, especially reducing confusion between classes. It mostly confuses Hate Speech with Offensive Language, because they are semantically close. DistilBERT’s errors are more semantically meaningful, while LSTM’s errors are more random.

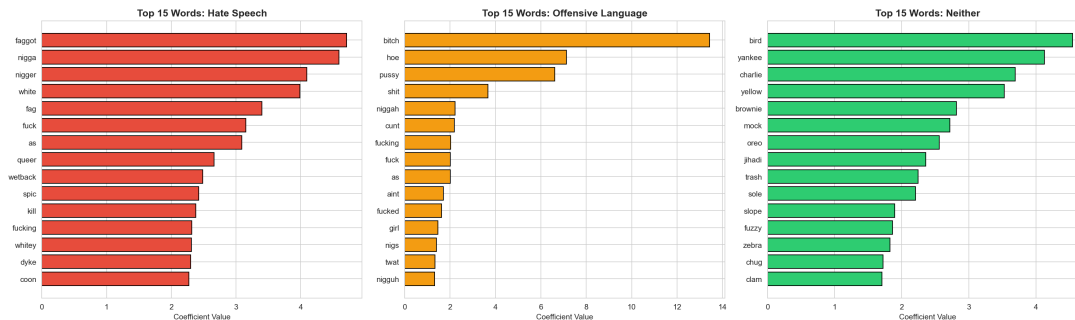


Figure 7: Top important words associated with each class according to the model.

These words provide insight into which features the model uses to differentiate between the categories.

3 Discussion

The results demonstrate that traditional machine learning models can achieve strong baseline performance for hate speech detection.

The evaluation results show that TF-IDF features do not consistently outperform Bag-of-Words representations but its performance varies across models. TF-IDF highlights important words while reducing the influence of very common terms. Deep learning models generally outperform classical approaches, with DistilBERT achieving the best overall performance. However, not all deep learning models consistently outperform traditional methods, as the Bidirectional LSTM shows slightly lower performance than the best classical model (SVM with Bag-of-Words) in terms of macro F1 score. DistilBERT achieved the best performance overall, with the highest accuracy (0.9161) and macro F1 score (0.7438), indicating strong performance across all classes. Classical models such as Random Forest and SVM performed competitively in terms of accuracy, but had lower macro F1 scores, suggesting weaker performance on minority classes. Naive Bayes performed the worst, particularly in handling class imbalance.

However, the confusion matrices indicate that distinguishing between hate speech and offensive language is still challenging. Both categories often share similar vocabulary, making it difficult for the models to clearly separate them.

Another limitation is the class imbalance in the dataset. Since offensive language appears more frequently than hate speech, the models may become biased toward predicting the majority class.

Future work could include techniques such as oversampling or class weighting to address the imbalance problem.

4 Conclusion

This study explored the use of traditional machine learning and deep learning models for detecting hate speech in tweets. Using TF-IDF feature extraction and several classification algorithms, strong baseline performance was achieved.

The results show that classical models such as Logistic Regression and Support Vector Machines perform well for this task but deep learning approaches such as DistilBERT outperform them. DistilBERT achieved the best results due to its ability to capture context and semantic meaning as it is pretrained on massive data and it is better at subtle language differences while LSTM learns from scratch and has limited context understanding because it relies on word sequences only.

However, distinguishing between hate speech and offensive language remains difficult due to vocabulary overlap and dataset imbalance.

Future work involving other deep learning approaches such as RoBERTa may further improve the ability of models to understand context and detect harmful language more accurately.

References

- [1] Thomas Davidson, Dana Warmley, Michael Macy, and Ingmar Weber. Hate speech and offensive language dataset. <https://www.kaggle.com/datasets/mrmorj/hate-speech-and-offensive-language-dataset>, 2017. Accessed: 2026-03-20.